

Wi-Fi can track more than heartbeats

In addition to heart rate monitoring, other health parameters can be detected using Wi-Fi signals.

- **Breathing rate detection:** Wi-Fi signals can capture subtle chest movements, allowing the system to track a person's breathing rate. Breathing patterns are more pronounced than heart rate, making detection easier and more reliable.
- **Apnea monitoring:** Apnea can be detected as a binary task. It is just checking whether a person is breathing.

Using simple Wi-Fi hardware with customised filters, the system can monitor breathing and identify apnea events without requiring any wearable sensors.

heart-rate prediction. It should rely on real data collected from participants, not simulations. The setup includes a transmitter-receiver pair for Wi-Fi communication. For data processing and analysis, the system can use Python or any other suitable programming language.

Even after preprocessing, the raw signal patterns remain difficult to interpret. Earlier methods relied on peak counting or simple heuristics, but these approaches break down when the user or the environment changes. A rule-based method may work in one setting but fail immediately in another. This is where ML becomes useful. An LSTM, designed for time-series data, generalises better across conditions and delivers stronger results than any heuristic method tested.

Start with a small model, so that it can run on low-cost hardware such as a Raspberry Pi, an ESP32, or an Adafruit board. This keeps computation light and makes the system easier to deploy. The hardware setup also shapes the design choices. Unlike previous research using multi-antenna systems to extract phase differences and angles of arrival, a single-antenna setup can now be used because standard devices like ESP32 and Raspberry Pi do not support advanced signal-engine functions. Multi-antenna cards, such as Intel 5300, are discontinued and difficult to access or program. Using an ESP32 and a

Raspberry Pi makes the approach more accessible and less dependent on specialised hardware.

ML provides flexibility beyond basic prediction. With larger datasets collected across environments or antennas, transfer learning becomes possible. A model trained on one setup can pass its learned weights to another system, improving performance without starting from scratch. This adaptability is challenging to achieve with heuristic approaches.

Experiments at 1m, 2m, and 3m may show that accuracy remains stable, whereas heuristic systems typically degrade with increasing distance. The ML model can adapt to new signal characteristics and reorganise its internal representation, enabling it to handle dynamic environments.

It remains to be seen whether touch-free heart-rate monitoring will become a mainstream health tool or remain confined to research settings. As datasets expand and transfer learning advances, these systems can become more flexible and reliable. With simpler, cost-effective hardware in the future, accuracy across different setups could improve, bringing us closer to practical, touch-free health monitoring that works anywhere. **EFY**

This story is inspired by the inputs from Nayan Sanjay Bhatia, PhD candidate at the University of California, Santa Cruz, who is also a co-author of the Pulse-Fi research paper. Co-author, Nidhi Agarwal is Senior Technology Journalist at EFY.

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